Part One: Foundations, Regression, and Generalized Linear Models

An Introduction to the Chapters and How They Fit Together

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1 What Part One Is

Quantitative Methods for Social and Behavioral Research with R is written for postgraduate students and applied researchers who read empirical studies, judge whether their conclusions hold, and increasingly conduct quantitative research themselves. It is not organized as a software-command manual or a specialist statistical reference. Questions, evidence, reasoning, and bounded answers come before procedures and output.

Part One contains fourteen numbered chapters. Chapters 2–6 build statistical and computational foundations. Chapters 7–9 develop the linear model; Chapters 10–12 extend regression to binary, categorical, and count outcomes; Chapter 13 focuses on prediction and validation; and Chapter 14 brings major threats to credibility to bear on complete quantitative arguments. The sequence therefore builds toward regression as its principal modelling family without treating every earlier method as regression.

What is assumed. Readers should have some introductory familiarity with statistics and algebra. Calculus and proof-based mathematical statistics are not assumed. Each chapter refreshes the notation and statistical ideas it actually needs at their point of use.

How to use this map. This introduction maps Chapters 1–8, the foundation and regression sequence developed here. Chapters 9–14 are the planned continuation described below. Chapter 1a is optional advanced depth and is never a prerequisite for a numbered chapter; the section *Chapter 1a: Optional Depth for Coordinated Studies* below describes what it adds and when to read it.

2 The Argument-Based Approach at a Glance

Chapter 1 develops the approach in full. This preview supplies only the map a reader needs to understand the sequence.

A quantitative study can be read as a **question–answer contract**. One question–answer pair is one argument, even when several arguments share data, records, or a fitted model. Each argument has linked **statistical** and **substantive** layers. Within each layer, documentary content performs four functions: Evidence and Justification support Inference, which supports a bounded Claim:

$$E + J \longrightarrow I \longrightarrow C.$$

The components are functions performed by material in the accessible record, not containers tied to particular section headings. Statistical content concerns observations, quantities, procedures, uncertainty, and checks. Substantive content situates those observations in the people or entities, conditions, setting, period, measurement meaning, and alternatives named by the question. Across layers, Evidence **situates**, Justification **augments and contextualizes**, Inference **interprets**, and Claim **bounds**. Reasons may be stated directly, located in related materials, or implicit but traceable; locating a reason does not establish that it is adequate.

The four functions in two layers give eight components, and the codes are used throughout Part One.

Table 1
The eight argument components.

	Evidence (E)	Justification (J)	Inference (I)	bounded Claim (C)
Statistical	**STAT-E** observations and how they were produced	**STAT-J** stated reasons for the comparison, procedure, and uncertainty	**STAT-I** estimand, result, uncertainty, and checks	**STAT-C** the statistical proposition and its conditions
Substantive	**SUB-E** those observations as people, conditions, setting, and period	**SUB-J** theory, measurement meaning, implementation, alternatives, and scope	**SUB-I** the result read in substantive units and context	**SUB-C** the question-facing answer and its bounds

One question can require several statistical estimands. They form parallel statistical strands within one argument only when they are jointly necessary for one integrated substantive answer. If the study owes separate answers, it owes separate question–answer pairs and arguments.

Where estimation or inference is involved, the **estimand** names the target quantity before the estimate is interpreted. A fixed-record descriptive summary may instead be a quantity known for the specified observations. Questions and claims can be descriptive, associational, predictive, or causal; the mode belongs to the question and supporting design, not to the software.

The approach has three ranked uses. **Reading** is primary: identify the pair, recover its component functions and documentary traces, and understand the reported bounded answer. **Evaluating quality** is a dependent secondary use that begins only after reading. **Constructing** an argument is an auxiliary use for researchers conducting a study. Chapter 1 introduces six quality-evaluation questions and a six-stage, two-phase construction workflow; later chapters use only the parts needed for their own reading problem. A reader records **not located** when support cannot be found and does not turn that documentary status into a claim that the work was absent.

The six quality-evaluation questions are **Question alignment**, **Evidence integrity**, **Justification adequacy**, **Inferential adequacy**, **Transition fidelity**, and **Claim calibration**. Each yields one of four outputs: *adequate for this purpose*, *needs revision*, *insufficient information to judge*, or *not applicable*. The question–answer pair as a whole then receives one of four conclusions: *supported as stated*, *supported only after narrowing or revision*, *not supported*, or *not assessable from the accessible record*. Chapters apply the questions their own reading problem needs, and they differ in how far they carry the verdicts: some grade every question with an output, others apply the questions in prose. *Not located* is a documentary status about the record, never an evaluation conclusion. Chapter 1a extends this apparatus for studies that owe several coordinated answers, adding a seventh question and a seventh workflow stage.

3 How Chapters 1–8 Progress

The first eight chapters make five moves:

1. establish a common method for reading quantitative studies;
2. develop data, description, sampling, and inferential foundations without requiring R;
3. make those foundations executable through plain, reproducible R;
4. build simple and multiple regression from the earlier ideas; and
5. coordinate several distinct questions without merging their answers.

Table 2
Chapters 1–8: responsibilities and dependencies.

Chapter	Main responsibility	How it connects
1. How to Read a Quantitative Study as an Argument	Connect the question, evidence, justification, statistical result, and bounded substantive answer; distinguish reading from quality evaluation	Supplies the common approach used throughout Part One
2. Understanding Data, Variation, and Sampling	Understand observations, variables, distributions, summaries, association, representation, and repeated-sampling variation	Supplies descriptive and sampling foundations for Chapters 3 and 5–7
3. Understanding Estimates, Uncertainty, and Statistical Tests	Understand estimands, estimates, standard errors, intervals, tests, assumptions, and bounded answers	Supplies the inferential foundations implemented in Chapter 6 and extended in regression
4. Reading Group Comparisons: Categorical Outcomes and Mean Outcomes	Read categorical and mean comparisons, omnibus tests, specific contrasts, and parallel strands serving one integrated answer	Supplies comparison structures for Chapter 6 and later indicator-variable models
5. Descriptive Statistics with R	Prepare and inspect records, calculate descriptive quantities, produce appropriate displays, and report what the specified observations show	Makes Chapter 2's descriptive reasoning executable and prepares the records used later

Chapter	Main responsibility	How it connects
6. Basic Inferential Statistics with R	Reproduce estimates, intervals, and tests; connect calculations to estimands, assumptions, and interpretations	Makes Chapters 3–4 executable and establishes the computational treatment of uncertainty used in regression
7. Reading a Simple Linear Regression	Develop one-predictor fitting, interpretation, uncertainty, diagnostics, and the distinction among fixed-record slopes, model-based slopes, and new-case predictions	Integrates Chapters 2–6 in a one-predictor setting and prepares multiple regression
8. Reading Multiple Regression Across Coordinated Questions	Develop adjustment, indicators, model comparison, fitted quantities, and prediction assessment while keeping separately owed answers distinct	Integrates the earlier foundations in an occupation study and transfers the reasoning to a randomized Project STAR application

4 Chapter 1a: Optional Depth for Coordinated Studies

Chapter 1a, *Extending the Argument-Based Approach to Quantitative Studies*, is optional advanced material. It is never a prerequisite for a numbered chapter, and every numbered chapter states the bounded answer it owes without it.

Chapter 1 develops the approach on studies that owe one answer at a time. Chapter 1a carries the same apparatus to a study that owes several answers at once, and extends it in three ways:

- it separates multiple question-level arguments from multiple statistical strands inside one argument, so coordinated questions are not collapsed into a single overstated answer;
- it adds a seventh quality-evaluation question, **Cross-question coherence**, for studies whose separately owed answers must be read together; and
- it extends the six-stage construction workflow to a seven-stage version and shows how regression instantiates question-specific estimands.

Its evidence is an upgraded version of the special-practice study from Chapter 1, now with 320 randomized participants, three distinct questions, imperfect attendance, missing delayed outcomes, adjustment, and interaction, followed by a transfer to a published study and its supporting materials.

When to read it. A reader who wants the full apparatus before meeting it in practice can continue directly from Chapter 1. Most readers will find it more useful alongside Chapter 8, where coordinated questions and separately owed answers first arise inside a fitted model.

Chapter 8 introduces the coherence question it needs at the point of use, so the sequence stays complete either way.

5 Conceptual and Computational Foundations

Chapters 2–4 and Chapters 5–6 form connected conceptual and computational foundations. The first group develops understanding through explanation, worked mathematics, and supplied results without requiring R. The second makes selected ideas executable through observation selection, transparent calculation, and interpretation. When an example returns, it returns to make a new step in the reasoning visible rather than merely repeating an earlier result.

Chapter 5 describes the records actually included. Chapter 6 introduces uncertainty about a stated estimand under sampling, assignment, or model assumptions. This is an analytical distinction, not a replacement for the four question modes: a descriptive population question may require statistical inference, while a regression can be descriptive, associational, predictive, or causal depending on the question and its justification. Running a calculation in R supplies no design warrant.

Three mathematical threads connect the chapters:

- covariance and correlation become a slope in Chapters 2, 5, and 7;
- standard errors and intervals for means become standard errors and intervals for a slope in Chapters 3, 6, and 7; and
- group comparisons become indicator-variable models in Chapters 4, 6, and 8.

The mathematics needed to understand a chapter's principal results remains in its numbered route. Fuller derivations, specialized comparisons, simulations, and computational variants may appear in signposted appendices.

6 What Follows After Chapter 8

The remaining chapters extend the regression sequence along the form of a relationship, the type of outcome, and the purpose of a model. Chapter 9 adds interaction and moderation. Chapters 10–12 address binary, nominal or ordinal, and count or rate outcomes. Chapter 13 develops predictive estimands, cross-validation, and regularization. Chapter 14 examines confounding structures, missingness, measurement problems, and selective reporting in a full credibility audit.

These chapters reuse the same reading discipline: name the question and estimand, interpret the result on a meaningful scale, attach uncertainty where appropriate, inspect the support and assumptions, and complete a bounded answer. More complex methods do not relax those obligations.

7 Routes and Commitments

A taught course can follow Chapters 1–8 in order, pairing Chapter 2 with Chapter 5 and Chapters 3–4 with Chapter 6 when the calendar allows. Chapters 7–8 are the regression destination toward which those foundations build. Chapter 1a is optional enrichment for such a course, better offered alongside Chapter 8 than as a second opening lesson.

A reader working alone should begin with Chapter 1 and may read Chapters 5 and 6 alongside Chapters 2–4 when learning through computation helps. Chapter 1a can wait until Chapter 8 makes its coordinated questions concrete.

A reader consulting one method should still read its opening question and closing bounded answer, not fitted output alone.

Across these routes, five commitments remain constant:

1. **Method follows question.** A procedure must help answer, qualify, or evaluate a declared question.
2. **Reading precedes evaluation.** The reader first records what the study reports, including reasons that may later prove inadequate.
3. **Interpretation does not repair support.** Substantive language cannot erase statistical, design, measurement, or scope limitations.
4. **The minimum computational route stays plain.** Chapters 5 onward use mostly base R and a few named packages. Optional extensions may use compact helper functions for repeated calculations.
5. **Required support remains in the numbered route.** Appendices may add depth, but the chapter's bounded answer must remain understandable without them.

Each numbered teaching route closes with a **Summary and Reading Bridge**, followed by unnumbered exercises. Optional appendices and references may follow.